1 divisibility and modular arithmetic

divisibility is one of the most basic properties of integers. an integer a ($a \neq 0$) can divide another integer b *if and only if* there is another integer c such that $b = ac$. we write that $a \mid b$ to say that a divides b, and $a \nmid b$ to say that a does not divide b. we call a the **factor** of b, and b the **multiple** of a.

formally, divisibility can be written as:

$$a \mid b \leftrightarrow \exists c \in \mathbb{Z}\,(b = ac)$$

1.1 properties of divisibility

1.1.1 additivity

theorem: if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

proof: by definition of divisibility, there must be some integer k and j such that $b = ak$ and $c = aj$. therefore, by substitution, $b + c = (ak + aj) = a(k + j)$. since both k and j are integers, $k + j$ is also just an integer, and by definition of divisibility, a is divisible by b where b is the product of a and $k + j$. ■

1.1.2 multiplication

theorem: if $a \mid b$, then $a \mid bc$ for all integers c.

proof: by definition of divisibility, there exist some integer k such that $b = ak$. therefore, $bc = (ak) \cdot c$. by the associativity of multiplication, $bc = a(kc)$. since k and c are integers, their product is also an integer. by definition of divisibility, a is divisible by an integer $bc = a(kc)$. ■

1.1.3 transitivity

theorem: if $a \mid b$ and $b \mid c$, then $a \mid c$.

proof: by definition of divisibility, there exist some integer k and j such that $b = ak$ and $c = bj$. therefore, by substitution, $c = (ak) \cdot j$. by definition of divisibility and associativity, we can see that $c = a(jk)$. ■

1.2 division "algorithm"

theorem: let a be an integer and let d be a positive integer. there are *unique* integers q and r, with $0 \leq r < d$, such that $a = dq + r$.

- a is called the **dividend**
- d is called the **divisor**
- q is called the **quotient**
- r is called the **remainder**
- $q = a \operatorname{div} d$ (integer division)

    - calculates the biggest possible integer quotient that can be divided from a number.
- $r = a \bmod d$ (modulo/remainder)
    - calculates the remainder (non-negative) after divided by the largest integer quotient possible.

1.3 property of modulo

theorem: if a and b are integers and m is a positive integer, we say that a *is congruent to* b modulo m *if and only if* $m \mid (a - b)$. we write this as $a \equiv b (\bmod m)$. we can write this theorem in a chain of equivalence:

$$a \equiv b (\bmod m) \Leftrightarrow a \bmod m = b \bmod m \Leftrightarrow m \mid (a - b)$$

this chain have three parts:

1. **congruency**; a is congruent to $b \bmod m$.
2. **remainders**; the remainder of a and b divided by m is the same.
3. **division**; $a - b$ is divisible by m.

in simple words, two integers a and b are congruent $\bmod m$ ($m > 0$) if and only if both numbers divided by m produce the exact same remainder. additionally, m should also evenly divide $a - b$.

1.4 properties of congruencies

1.4.1 algebraic form

theorem: if a is congruent to $b (\bmod m)$ ($a \equiv b (\bmod m)$), then there exists some integer k such that $a = b + km$.

this is useful for proofs because it lets us replace the congruence symbol ($\equiv$) with an equal ($=$) sign.

1.4.2 addition and multiplication

theorem: let m be a positive integer. if $a \equiv b (\bmod m)$ and $c \equiv d (\bmod m)$, then $(a + c) \equiv (b + d)(\bmod m)$ and $ac \equiv bd (\bmod m)$.

in simpler words: if a and b are congruent $(\bmod m)$, and c and d are also congruent $(\bmod m)$, then $(a + c)$ is congruent to $(b + d)(\bmod m)$ and $a \cdot c$ is congruent to $b \cdot d (\bmod m)$.

proof (addition): assume $a \equiv b (\bmod m)$ and $c \equiv d (\bmod m)$. according to the property of modulo, this means $m \mid (a - b)$ and $m \mid (c - d)$. by the definition of divisibility, this means that there exists some integers k and j such that $a - b = km$ and $c - d = jm$. since we want to prove $(a + c) \equiv (b + d)(\bmod m)$; by doing a bit of algebra, we can get:

$$\begin{aligned}
&(a+c)-(b+d) \\
&= (a-b)+(c-d) \\
&= km+jm \qquad \text{(substitution)} \\
&= m(j+k)
\end{aligned}$$

therefore, $m \mid ((a+c)-(b+d))$, which means $(a+c) \equiv (b+d) (\operatorname{mod} m)$. ■

proof (multiplication): we can use the algebraic form of congruence from the theorem above: $a = b + km$ and $c = d + jm$ for some integers k and j. substitute these into the product of $a \cdot c$:

$$\begin{aligned}
ac &= (b+km)(d+jm) \\
&= bd + bjm + dkm + kmjm \\
&= bd + m(bj + dk + kmj) \\
ac - bd &= m(bj + dk + kmj)
\end{aligned}$$

since $(bj + dk + kjm)$ is an integer, $m \mid (ac - bd)$, which means $ac \equiv bd (\operatorname{mod} m)$.

1.5 applications of congruencies

congruencies have many applications within computer science; we will look at three of them in this section.

1.5.1 hash functions

hash functions allow us to quickly and efficiently locate stored data. we apply a hash function that determines the storage location of the record based on its id. a common hash function is $h(k) = k \operatorname{mod} n$, where n is the number of available storage locations.

1.5.2 prng

congruencies can be used to generate seemingly random sequences. we can create our own simple pseudorandom number generator in two steps: first, we choose some constants: a modulus m, a multiplier a, an increment c, and a seed x_0.

then, each number in the sequence is decided by this formula: $x_{n+1} = (ax_n + c) \operatorname{mod} m$. here's an example where we have $m = 9, a = 7, c = 4, x_0 = 3$:

- $x_1 = (7x_0 + 4) \operatorname{mod} 9 = 25 \operatorname{mod} 9 = 7$
- $x_2 = (7x_1 + 4) \operatorname{mod} 9 = 53 \operatorname{mod} 9 = 8$
- $x_3 = (7x_2 + 4) \operatorname{mod} 9 = 60 \operatorname{mod} 9 = 6$
- $x_4 = (7x_3 + 4) \operatorname{mod} 9 = 46 \operatorname{mod} 9 = 1$
- $x_5 = (7x_4 + 4) \operatorname{mod} 9 = 11 \operatorname{mod} 9 = 2$

1.5.3 cryptography

cryptography is the study of "secret messages"; and one of the earliest forms of encryption, the caesar cipher, is based on congruency. to encode a message using the caesar cipher, we first choose a shift index s. then, we map each letter in the alphabet to a number; and finally, for each letter, we calculate $f(p) = (p + s) \bmod 26$ (assuming the english alphabet, with 26 letters).

for example, we choose the shift index $s = 9$. we want to encrypt this secret message "ATTACK". if A-Z is mapped 0-25, the message would be 019190210. then, we apply the function to each number: $f(0) = 9$, $f(19) = 2$, $f(2) = 11$, $f(10) = 19$. at the end, we would get this encrypted message: 92291119. to an attacker, even if they are smart enough to figure out we are mapping each number to the alphabet, without the shift index s, the message will be meaningless; since 92291119 maps to "JCCJLT".

to decrypt the message, we use the inverse function: $f^{-1} = (p - s) \bmod 26$ (again, assuming the alphabet have 26 letters).